

ZapBox Simple – Operating Instructions

Language: English | **Version:** oi967242

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Overview

The **ZapBox Compact** is an electronic switch for Bitcoin Lightning payments. A payment via the Lightning Network can be used to switch an output – ideal for vending machines, presentations, event control, and many other applications.

Basic Equipment

Component	Description
Microcontroller	T-Display-S3 with 1.9" LCD display
Front panel	35° display
Input	Dual USB-A and USB-C
Output	Dual USB-A and USB-C
Control	Two on-board micro switches
Option BTC ticker	Activatable via the Web Installer

Views

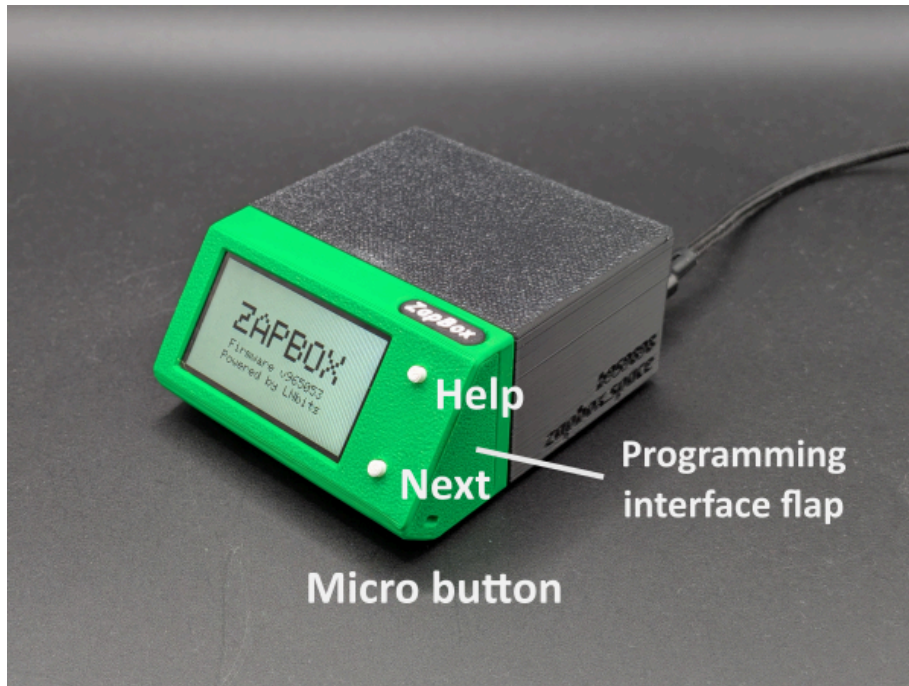


Figure 1: Front view



Figure 2: Rear view

Connections

Input - Dual USB-A and USB-C socket for power supply (5V)

Power the device via the **Power IN** connector using a USB-C/A cable with **5 V DC (max. 5 A)**.

Note: The USB power connector does not support automatic USB-C power negotiation (no USB-C Power Delivery). Some USB-C chargers or power modules therefore do not recognize the ZapBox as a load and will not supply power. In this case, use a **USB-A output** of the power supply or an alternative 5 V power source. The maximum current must not exceed 3 A.

Input - USB-C socket on the microcontroller (data access, behind the right side panel)

To read or transfer data from the device, connect the ZapBox to a computer or laptop:

1. On the **right side of the front panel** there is a small, concealed flap.
2. Open the flap by pushing it to the right from below using a **narrow screwdriver**.
3. Connect a USB-C cable to the microcontroller's connector underneath.



Figure: Opening the panel and USB-C connector for data

Important note: The USB connector directly on the microcontroller is intended exclusively for flashing the firmware and transferring configuration parameters. During the flashing process, no load may be connected or switched at the output, as this can lead to malfunctions or **damage to the microcontroller**.

It is therefore recommended to either:

- not connect any load at the output while the USB connection is active, or
- additionally connect the regular **Power-IN input** to the same power supply. This ensures that the current for the power relay does not flow through the microcontroller and overload it.

Output - Dual USB-A and USB-C (switched 5V voltage)

The USB sockets are switched via a relay contact. The **total load** of the sockets should **not exceed 3 A**.

Controls

The ZapBox has two small **on-board micro switches** that are connected directly to the microcontroller. All functions can be accessed via the micro switches. In addition, the ZapBox has a reset button on the underside of the front panel and two slide switches on the side.

Function Overview - Micro Switches

Function	Micro switch
Show help page	Press HELP once
Next page / product change	Press NEXT once
Show REPORT page	Press HELP twice
Open config mode	Hold NEXT for at least 5 sec.

Setup and Commissioning

The ZapBox is tested after manufacturing and shipped with the current firmware - however, it is not yet parameterized. The software is under active development, so it is recommended to flash the ZapBox with the latest firmware right at the start and then perform the parameterization. A convenient **Web Installer** is available for this.

Here is a step-by-step guide for the setup:

1. Open the right side panel of the front panel, as described above under "Input - USB-C socket on the microcontroller".
2. Connect the ZapBox to the USB-C port with a cable and connect it to a computer.
3. Open a Chromium browser, for example Google Chrome, Microsoft Edge, Brave, Vivaldi, Opera, or [Helium](#).
4. Go to the Web Installer page <https://installer.zapbox.space/>.
5. Flash the latest "Latest" version, as described under point 1 of the Web Installer.
6. After the flashing process, close the small window and go to point 3 - Load config values. There, select the  **Connect** button.
7. You should now see  **Connected** and  **Config mode** in the green field, provided the ZapBox is now in **SERIAL CONFIG MODE**. The display should show this. If not, check point 2 - Prepare connection.
8. The ZapBox needs three parameters: **WiFi SSID** / **WiFi password** / **Device settings string**. You get the device settings string from your LNbits wallet. To do this, add the **Bitcoin Switch** or **ZapBox** extension. The ZapBox extension also supports the NFC module, otherwise they are identical.
9. After all three parameters have been entered, save them once with the  **Write Config** button and restart the ZapBox with the  **Restart** button.

That should be it. After initialization, the ZapBox will display the product's QR code and is ready for the first payment and subsequent action at the USB output.

In case of errors or malfunctions, please refer to the "Error Detection & Report" and "Troubleshoot" chapters further down on the Web Installer page.

Technical Data

Property	Value
Supply voltage	5 V DC via USB-C
Maximum input current	5.0 A

Property	Value
Output power	max. 3.0 A (recommended)
Display	1.9" LCD (T-Display-S3)
Communication	Wi-Fi (ESP32-S3)
Payment protocol	Bitcoin Lightning Network

Safety Instructions

- Operate the device only with the specified supply voltage.
- Do not exceed the maximum current load of the outputs.
- Do not perform any work on the relay contacts under load.
- The device is not suitable for use in damp or wet environments.
- Keep out of the reach of children.

Further Links

Resource	Link
Overview of all ZapBox models	https://zapbox.space/
Web Installer, quick overview & troubleshooting	https://installer.zapbox.space/
Detailed documentation (parameters & functions)	https://ereignishorizont.xyz/zapbox/
GitHub repository (software, PCB layouts, 3D print files, operating instructions)	https://github.com/AxelHamburch/ZapBox
ZapBox Extension	https://github.com/AxelHamburch/zapbox_extension
LNbits	https://lnbits.com/

Subject to changes and errors. As of: 2026